

TermPlan-MT: Graph-Grounded French–English Translation for SmPC Section 4.8 (Results Snapshot)

Project repository snapshot

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Abstract

We summarise **TermPlan-MT**, a French-to-English pipeline for regulatory SmPC §4.8 text (Keytruda slice, 127 aligned sentence pairs). Named entities in French are grounded in a **MedDRA** graph (Neo4j), terminology is **planned** into global locks, and six MT conditions (S1–S5 plus a Mistral S5 branch) are scored with **sacreBLEU** BLEU/chrF++, hierarchy-aware **HTM**, a new **hypothesis–reference HTM agreement** score, dataset **rHTM**, and span **CCR**. We report two NER regimes: prompted BioMistral vs. fine-tuned BioMistral (Unsloth), and list ten French NER surfaces that trigger **ambiguous** string grounding for manual ontology review.

1 Task and setup

The goal is terminology-consistent translation under a controlled vocabulary: French clinical phrases are linked to MedDRA concepts before and after decoding. Evaluation uses the same segment JSONL for NER spans as for CCR/HTM. Neo4j must be running for graph metrics.

2 Automatic metrics (short definitions)

- **BLEU / chrF++**: Corpus scores with **sacreBLEU** over all aligned segment hypotheses vs. a single English reference per segment (`pipeline/metrics/corpus_scores.py`). Document variants **BLdoc** (macro mean per document) and **BLcn** (BLEU on space-joined segments per document) appear in `scores_summary.csv`.
- **HTM**: For each unique French `terms[].word` that grounds in Neo4j, we look for MedDRA-aligned English renderings in the system **hypothesis** and assign 1.0 / 0.5 / 0.0 using level and branch checks (`compute_htm`).
- **rHTM (htm_en_ref_dataset)**: Same HTM machinery on the human **en_ref** string, averaged once per dataset (ceiling for MedDRA wording in the reference).
- **htm_hyp_ref_agreement**: Per system, over **grounded** French spans only: HTM-style score on **hyp** and the same score on **en_ref**; we report the mean of $1 - |s_{\text{hyp}} - s_{\text{ref}}|$. High values mean the model’s ontology-aligned rendering tier tracks the human sentence for the same French cue.
- **CCR**: Fraction of extracted NER spans that receive a non-null MedDRA grounding (dataset level).

3 Main numeric tables

Numbers are copied from `results/*/figures/scores_summary.csv` at export time (two decimal places; COMET omitted when not computed).

System	BLEU	chrF	HTM	HypRefAg	CCR
<i>NER = prompted BioMistral (segments_ner_biollm.jsonl)</i>					
S1 NLLB	18.89	31.90	0.11	0.68	0.35
S2 Mistral (doc)	19.84	35.58	0.20	0.66	0.35
S3 GraphRAG	10.79	32.65	0.25	0.59	0.35
S4 rerank	10.87	32.85	0.24	0.60	0.35
S5 NLLB + boost	17.34	30.55	0.13	0.66	0.35
S5 Mistral + boost	10.82	32.64	0.27	0.58	0.35
<i>NER = fine-tuned BioMistral (segments_ner_unsloth_full.jsonl)</i>					
S1 NLLB	18.89	31.90	0.17	0.42	0.32
S2 Mistral (doc)	20.37	35.85	0.25	0.49	0.32
S3 GraphRAG	15.36	34.60	0.26	0.42	0.32
S4 rerank	15.66	34.92	0.26	0.43	0.32
S5 NLLB + boost	17.54	31.02	0.17	0.39	0.32
S5 Mistral + boost	15.98	34.85	0.26	0.43	0.32

Table 1: Fluency (BLEU/chrF), HTM on hypotheses, hypothesis–reference agreement (HypRefAg), and dataset CCR. rHTM (not shown) is ≈ 0.13 (prompt) / ≈ 0.16 (FT) on this export.

Takeaway. Fine-tuned NER raises **HTM** on several systems (more audited spans align with MedDRA renderings in **hyp**) but lowers **HypRefAg**: the model’s hierarchy scores diverge slightly more from the human reference tier than under prompted NER, at similar BLEU for S1. Graph-heavy lines (S3–S5) remain fluency-expensive vs. S1/S2.

4 Ambiguous French surfaces (manual follow-up)

String grounding warns when one normalised French key matches **multiple** MedDRA concepts. We export ten (**segment_id**, **term**) examples for prompted NER in `error_analysis/ambiguous_spans_top10.md` (and `.csv`), generated by `tools/list_ambiguous_spans.py`. Use the official MedDRA browser to inspect competing PT/LLT branches for each surface.

5 Reproducibility commands

```
docker compose up -d # Neo4j
export PYTHONPATH=.
python tools/evaluate.py --no-comet \
  --results-dir results/ner_biollm \
  --segments data/section48/segments_ner_biollm.jsonl
python tools/plot_figures.py --results-dir results/ner_biollm \
  --segments data/section48/segments_ner_biollm.jsonl
python tools/list_ambiguous_spans.py --n 10
```

Omit `--no-comet` to compute COMET when the Unbabel model is installed (slower).

6 What was added in code (this milestone)

`compute_htm_hyp_vs_ref` in `pipeline/metrics/htm.py`; column `htm_hyp_ref_agreement` in `collect_system_metrics/scores_summary.csv`; console column `HypRefAg` plus stderr legend in `tools/evaluate.py`; optional `--no-comet` for faster re-scoring; `tools/list_ambiguous_spans.py` for manual ambiguity review; README and docs/CONCLUSIONS.md updated accordingly.